BRAF fusion-hotspot benchmark report

Study: msk_impact_2017

Abstract

This report analyzes BRAF gene fusions from the msk_impact_2017 study, covering 41 fusion events. 33/33 fusions with known frame status (100.0%) were in-frame. 33/40 fusions (82.5%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.00060718, statistically significant at alpha=0.05). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 100.0% in-frame and 82.5% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 41 analyzed fusion events, identifying 25 distinct fusion partner genes. The most frequent partner was SND1, observed in 8 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 31/33 (93.9%) of in-frame fusions retained the domain, p=0.00060718 (statistically significant at alpha=0.05). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.001998.

Domain disruption

Disruption of the RAS-binding domain and Cysteine-rich domain was tested with Fisher's exact test comparing in-frame fusions against all others; 31/33 (93.9%) of in-frame fusions disrupted the domain, p=0.0049969 (statistically significant at alpha=0.05). The RAS-binding domain and Cysteine-rich domain appear to require loss or disruption rather than retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Junction concentration

50% of 34 fusions with this gene as the 3' partner join at exon 9, against 4% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.000999001, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 40 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 380 aa (exon 8; label-permutation p=0.016983). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -78 aa from the nearest configured domain boundary at 458 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=33) and "not_retained" (n=7). For Frame_status, retained: 31/31 (100.0%, 95% CI 89.0%-100.0%); not_retained: 2/2 (100.0%, 95% CI 34.2%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of 48.303 vs 30.8571, p=0.273788 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (25 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for BRAF in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (18 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 5 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run (2 row(s) in retention_counter_intuitive_events, 2 row(s) in disruption_counter_intuitive_events); see the accompanying table.

Mutation cooccurrence

Mutation cooccurrence produced results for this run (1 row(s) in cooccurrence_results); see the accompanying table.

Window detection

Window detection produced results for this run (46 row(s) in window_scan, 23 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SND1	8	0.1951219512195122	0.26994040818153375	0.30004340774793187	0.8777027027027027	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.37929201596853973	1
KIAA1549	4	0.0975609756097561	0.26994040818153375	0.30004340774793187	0.9162162162162162	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3558007503126399	2
AGAP3	2	0.04878048780487805	0.26994040818153375	0.30004340774793187	0.9945945945945946	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3529233607279333	3
LINC00244	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	1.0	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3464170983680124	4
CCDC6	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9945945945945946	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3456062875572016	5
CUL1	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9945945945945946	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3456062875572016	6
FAM131B	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9945945945945946	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3456062875572016	7
METTL2B	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9945945945945946	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3456062875572016	8
PHTF2	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9945945945945946	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3456062875572016	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PRKAR1B	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9945945945945946	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3456062875572016	10
GIPC2	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9297297297297298	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.33587655782747183	11
KDM7A	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9297297297297298	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.33587655782747183	12
OSBPL9	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9297297297297298	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.33587655782747183	13
PRKAR2B	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9297297297297298	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.33587655782747183	14
ZNF207	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.9297297297297298	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.33587655782747183	15
CDK5RAP2	2	0.04878048780487805	0.26994040818153375	0.30004340774793187	0.8378378378378378	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3294098472144198	16
AGK	3	0.07317073170731707	0.26994040818153375	0.30004340774793187	0.7135135135135136	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3180782717365028	17
MRPS33	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.7135135135135136	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.3034441253950394	18
PJA2	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.6864864864864865	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.29939007134098533	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RBM33	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.681081081081081	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.29857926053017453	20
SCRIB	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.681081081081081	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.29857926053017453	21
LUC7L2	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.4972972972972973	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.27101169296260696	22
MKRN1	3	0.07317073170731707	0.26994040818153375	0.30004340774793187	0.23603603603603596	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2464566501148812	23
PARP12	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	None	0.6160624160231085	confidence_certainty, domain_disruption, domain_retention, recurrence	0.23107893925648515	24
VIPR2	1	0.024390243902439025	0.26994040818153375	0.30004340774793187	0.0	0.6160624160231085	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.19641709836801238	25

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
287	0	33	1	6	0.0	0.17500000000000004	0.7569619513137055
327	3	30	2	5	0.25	0.20399751978699343	0.6903751127180751
380	3	30	4	3	0.075	0.01084985914707277	1.9645758997920526
381	20	13	4	3	1.1538461538461537	1.0	-0.0
393	25	8	4	3	2.34375	0.36939377457953293	0.43251042803096634
438	25	8	5	2	1.25	1.0	-0.0
439	33	0	5	2	None	0.026923076923076925	1.569875307956561
565	33	0	6	1	None	0.175	0.7569619513137056

domain_disruption tables

frame_domain_contingency_table

31	3
2	4

frame_domain_contingency_tables_by_tumor_type

stratum	table
ACRM	[1, 1], [0, 0]
AGNG	[1, 0], [0, 0]
CHOL	[1, 0], [0, 0]
COAD	[2, 0], [0, 0]
CUP	[1, 0], [0, 0]
DDLS	[1, 0], [0, 0]
ILC	[1, 0], [0, 0]
LUAD	[4, 0], [1, 1]
MFH	[1, 0], [0, 0]
PAAC	[2, 0], [1, 0]
PAAD	[3, 1], [0, 1]
PAST	[2, 0], [0, 0]
PRAD	[3, 0], [0, 1]
PRSCC	[1, 0], [0, 0]
PXA	[1, 0], [0, 0]
SKCM	[3, 0], [0, 0]
THAP	[1, 0], [0, 0]
THPA	[1, 0], [0, 0]
THPD	[1, 0], [0, 0]
VMGCT	[0, 1], [0, 0]
VMM	[0, 0], [0, 1]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	40	6	0	34	0.0
cysteine_rich	Cysteine-rich domain	40	6	0	34	0.0

domain_retention tables

frame_domain_contingency_table

31	2
2	5

frame_domain_contingency_tables_by_tumor_type

stratum	table
ACRM	[1, 1], [0, 0]
AGNG	[1, 0], [0, 0]
CHOL	[1, 0], [0, 0]
COAD	[2, 0], [0, 0]
CUP	[1, 0], [0, 0]
DDLS	[1, 0], [0, 0]
ILC	[1, 0], [0, 0]
LUAD	[4, 0], [1, 1]
MFH	[1, 0], [0, 0]
PAAC	[2, 0], [1, 0]
PAAD	[3, 1], [0, 1]
PAST	[2, 0], [0, 0]
PRAD	[3, 0], [0, 1]
PRSCC	[1, 0], [0, 0]
PXA	[1, 0], [0, 0]
SKCM	[3, 0], [0, 0]
THAP	[1, 0], [0, 0]
THPA	[1, 0], [0, 0]
THPD	[1, 0], [0, 0]
VMGCT	[0, 0], [0, 1]
VMM	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	40	33	1	6	0.06050420168067227

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	34	41	0.8292682926829268

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
AGAP3	2	2	2	2
AGK	3	3	3	3
CCDC6	1	1	1	1
CDK5RAP2	2	2	2	2
CUL1	1	1	1	1
FAM131B	1	1	1	1
GIPC2	1	1	1	1
KDM7A	1	1	1	1
KIAA1549	4	4	4	4
LINC00244	1	1	1	1
LUC7L2	1	1	1	1
METTL2B	1	1	1	1
MKRN1	3	3	3	3
MRPS33	1	1	1	1
OSBPL9	1	1	1	1
PARP12	1	1	1	1
PHTF2	1	1	1	1
PJA2	1	1	1	1
PRKAR1B	1	1	1	1
PRKAR2B	1	1	1	1
RBM33	1	1	1	1
SCRIB	1	1	1	1
SND1	8	8	8	8
VIPR2	1	1	1	1
ZNF207	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
7	140482000	140483000	7	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0007392-T01-IM5-39, EVT-P-0007533-T01-IM5-40, ... (7 total)
7	140489000	140490000	4	EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0008140-T01-IM5-42, EVT-P-0008789-T01-IM5-36
7	140491000	140492000	4	EVT-P-0002305-T01-IM3-31, EVT-P-0004203-T01-IM5-23, EVT-P-0005121-T02-IM5-28, EVT-P-0009694-T01-IM5-37
7	140493000	140494000	4	EVT-P-0004795-T01-IM5-13, EVT-P-0009024-T01-IM5-32, EVT-P-0009757-T01-IM5-1, EVT-P-0012395-T01-IM5-47
7	140488000	140489000	3	EVT-P-0002197-T01-IM3-44, EVT-P-0005834-T01-IM5-22, EVT-P-0009097-T01-IM5-26
7	140481000	140482000	2	EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24
7	140484000	140485000	2	EVT-P-0010398-T01-IM5-18, EVT-P-0011111-T01-IM5-25
7	140487000	140488000	2	EVT-P-0004032-T01-IM5-27, EVT-P-0010494-T01-IM5-15
7	140490000	140491000	2	EVT-P-0001687-T01-IM3-41, EVT-P-0007861-T01-IM5-2
7	140497000	140498000	2	EVT-P-0005553-T02-IM5-17, EVT-P-0010945-T01-IM5-5
7	140434000	140435000	1	EVT-P-0004373-T02-IM5-34
7	140454000	140455000	1	EVT-P-0002516-T01-IM3-21
7	140485000	140486000	1	EVT-P-0008920-T01-IM5-48
7	140486000	140487000	1	EVT-P-0008567-T01-IM5-38
7	140492000	140493000	1	EVT-P-0004871-T01-IM5-16
7	140495000	140496000	1	EVT-P-0004477-T01-IM5-3
7	140496000	140497000	1	EVT-P-0000913-T02-IM5-4
7	140501000	140502000	1	EVT-P-0006412-T01-IM5-30

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
381	17	17	5979	False
439	8	8	1133	False

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
393	5	5	2488	False
327	4	4	2205	False
380	2	2	4081	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	9	17	34	0.5	0.0355254721839757	15.524535168109496
three_prime	11	8	34	0.23529411764705882	0.00759383244169303	9.773682587431338
three_prime	10	5	34	0.14705882352941177	0.023790504819061833	2.9236005763253408
three_prime	8	3	34	0.08823529411764706	0.03181526754464332 5	1.0317898466803144
five_prime	8	2	6	0.3333333333333333	0.036171867609808395	1.7493720516314597
five_prime	6	1	6	0.16666666666666666	0.005670411906288429	1.4743877545518123
five_prime	7	1	6	0.16666666666666666	0.031605057650063585	0.7563078614453926
five_prime	10	1	6	0.16666666666666666	0.007693682141618406	1.3440623394036564
five_prime	13	1	6	0.16666666666666666	0.12010867851549772	0.27088388827772564
three_prime	18	1	34	0.029411764705882353	0.02802623419484355	0.20789716208808656

mechanistic_interpretation tables

retention_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0003502-T01-IM5-19	P-0003502-T01-IM5	PJA2	438	lost
EVT-P-0002197-T01-IM3-44	P-0002197-T01-IM3	SND1	380	lost

disruption_counter_intuitive_events

event_id	sample_id	partner_gene	breakpoint_protein_position	domain_status
EVT-P-0003502-T01-IM5-19	P-0003502-T01-IM5	PJA2	438	retained
EVT-P-0002197-T01-IM3-44	P-0002197-T01-IM3	SND1	380	retained

mutation_cooccurrence tables

cooccurrence_results

comparator_gene	alteration_type	protein_change	label	contingency_table	odds_ratio	p_value	direction	cohort_sample_count	supplied_cohort_sample_count	excluded_cohort_sample_count	excluded_fusion_positive_sample_count	excluded_comparator_sample_count	counting_unit	eligibility_source	eligible_sample_ids_sha256	fusion_positive_sample_count	comparator_altered_sample_count	both_altered_sample_count
BRAF	point_mutation	V600E	BRAF point_mutation (V600E)	[1, 40], [294, 10610]	0.902210884353 7415	1.0	mutually_exclusive	10945	10945	0	0	0	sample	joint_assay_metadata	852528ceb010d68c171a546820e23558b07b40ad14d0de1e9421b126cb1f1a68	41	295	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
127	327	200	5	35	3	30	0.25	0.20399751978699343	0.6903751127180751	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0006412-T01-IM5-30, EVT-P-0010945-T01-IM5-5
180	380	200	7	33	3	30	0.075	0.01084985914707277	1.9645758997920526	EVT-P-0000913-T02-IM5-4, EVT-P-0002197-T01-IM3-44, EVT-P-0004477-T01-IM5-3, EVT-P-0004871-T01-IM5-16, EVT-P-0005553-T02-IM5-17, ... (7 total)
181	381	200	24	16	20	13	1.1538461538461537	1.0	-0.0	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (24 total)
193	393	200	29	11	25	8	2.34375	0.36939377457953293	0.43251042803096634	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (29 total)
227	327	100	5	35	3	30	0.25	0.20399751978699343	0.6903751127180751	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0006412-T01-IM5-30, EVT-P-0010945-T01-IM5-5
238	438	200	30	10	25	8	1.25	1.0	-0.0	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (30 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
277	327	50	5	35	3	30	0.25	0.20399751978699343	0.6903751127180751	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0006412-T01-IM5-30, EVT-P-0010945-T01-IM5-5
280	380	100	7	33	3	30	0.075	0.01084985914707277	1.9645758997920526	EVT-P-0000913-T02-IM5-4, EVT-P-0002197-T01-IM3-44, EVT-P-0004477-T01-IM5-3, EVT-P-0004871-T01-IM5-16, EVT-P-0005553-T02-IM5-17, ... (7 total)
281	381	100	24	16	20	13	1.1538461538461537	1.0	-0.0	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (24 total)
287	337	50	5	35	3	30	0.25	0.20399751978699343	0.6903751127180751	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0006412-T01-IM5-30, EVT-P-0010945-T01-IM5-5
287	387	100	24	16	20	13	1.1538461538461537	1.0	-0.0	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (24 total)
293	393	100	28	12	25	8	4.166666666666667	0.16797757509831812	0.7747486924473793	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (28 total)
302	327	25	4	36	3	30	0.6	0.5522486048801838	0.2578653725929749	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0010945-T01-IM5-5
327	352	25	4	36	3	30	0.6	0.5522486048801838	0.2578653725929749	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0010945-T01-IM5-5
327	377	50	4	36	3	30	0.6	0.5522486048801838	0.2578653725929749	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0010945-T01-IM5-5
327	427	100	28	12	25	8	4.166666666666667	0.16797757509831812	0.7747486924473793	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (28 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
331	381	50	19	21	17	16	2.65625	0.4124220374220374	0.3846581366109272	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (19 total)
338	438	100	25	15	22	11	2.6666666666666665	0.3920474415830453	0.4066613759105142	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (25 total)
339	439	100	33	7	30	3	13.333333333333334	0.010849859147072769	1.9645758997920526	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (33 total)
343	393	50	24	16	22	11	5.0	0.09415068795873749	1.0261765022475087	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (24 total)
356	381	25	19	21	17	16	2.65625	0.4124220374220374	0.3846581366109272	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (19 total)
365	565	200	34	6	30	3	7.5	0.05474705474705476	1.2616392398030527	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (34 total)
368	393	25	24	16	22	11	5.0	0.09415068795873749	1.0261765022475087	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (24 total)
380	405	25	24	16	22	11	5.0	0.09415068795873749	1.0261765022475087	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (24 total)
380	430	50	24	16	22	11	5.0	0.09415068795873749	1.0261765022475087	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (24 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
380	480	100	33	7	30	3	13.333333333333334	0.010849859147072769	1.9645758997920526	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (33 total)
380	580	200	34	6	30	3	7.5	0.05474705474705476	1.2616392398030527	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (34 total)
381	406	25	22	18	22	11	None	0.0017069701280227592	2.767774078972199	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002305-T01-IM3-31, EVT-P-0004032-T01-IM5-27, ... (22 total)
381	431	50	22	18	22	11	None	0.0017069701280227592	2.767774078972199	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002305-T01-IM3-31, EVT-P-0004032-T01-IM5-27, ... (22 total)
381	481	100	31	9	30	3	60.0	0.00014160385677413535	3.8489249178827287	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002305-T01-IM3-31, ... (31 total)
381	581	200	32	8	30	3	25.0	0.0015383328076826529	2.812949697756797	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002305-T01-IM3-31, ... (32 total)
388	438	50	6	34	5	28	1.0714285714285714	1.0	-0.0	EVT-P-0003502-T01-IM5-19, EVT-P-0008567-T01-IM5-38, EVT-P-0008920-T01-IM5-48, EVT-P-0010398-T01-IM5-18, EVT-P-0010494-T01-IM5-15, ... (6 total)
389	439	50	14	26	13	20	3.9	0.3870387415279055	0.41224556110594196	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (14 total)
393	418	25	5	35	5	28	None	0.564686143633512	0.24819286892566125	EVT-P-0008567-T01-IM5-38, EVT-P-0008920-T01-IM5-48, EVT-P-0010398-T01-IM5-18, EVT-P-0010494-T01-IM5-15, EVT-P-0011111-T01-IM5-25

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
393	443	50	14	26	13	20	3.9	0.3870387415279055	0.41224556110594196	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (14 total)
393	493	100	14	26	13	20	3.9	0.3870387415279055	0.41224556110594196	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (14 total)
393	593	200	15	25	13	20	1.625	0.69127409142889	0.16034971992983757	EVT-P-0001036-T01-IM3-33, EVT-P-0002516-T01-IM3-21, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, ... (15 total)
414	439	25	9	31	8	25	1.92	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (9 total)
438	463	25	9	31	8	25	1.92	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (9 total)
438	488	50	9	31	8	25	1.92	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (9 total)
438	538	100	9	31	8	25	1.92	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (9 total)
438	638	200	10	30	8	25	0.8	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0002516-T01-IM3-21, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, ... (10 total)
439	464	25	8	32	8	25	None	0.3087122845636777	0.5104460883278611	EVT-P-0001036-T01-IM3-33, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, EVT-P-0007392-T01-IM5-39, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
439	489	50	8	32	8	25	None	0.3087122845636777	0.5104460883278611	EVT-P-0001036-T01-IM3-33, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, EVT-P-0007392-T01-IM5-39, ... (8 total)
439	539	100	8	32	8	25	None	0.3087122845636777	0.5104460883278611	EVT-P-0001036-T01-IM3-33, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, EVT-P-0007392-T01-IM5-39, ... (8 total)
439	639	200	9	31	8	25	1.92	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0002516-T01-IM3-21, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (9 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
381	481	100	31	9	30	3	60.0	0.00014160385677413535	3.8489249178827287	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002305-T01-IM3-31, ... (31 total)
381	581	200	32	8	30	3	25.0	0.0015383328076826529	2.812949697756797	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002305-T01-IM3-31, ... (32 total)
381	406	25	22	18	22	11	None	0.0017069701280227592	2.767774078972199	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002305-T01-IM3-31, EVT-P-0004032-T01-IM5-27, ... (22 total)
280	380	100	7	33	3	30	0.075	0.01084985914707277	1.9645758997920526	EVT-P-0000913-T02-IM5-4, EVT-P-0002197-T01-IM3-44, EVT-P-0004477-T01-IM5-3, EVT-P-0004871-T01-IM5-16, EVT-P-0005553-T02-IM5-17, ... (7 total)
339	439	100	33	7	30	3	13.333333333333334	0.010849859147072769	1.9645758997920526	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (33 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
365	565	200	34	6	30	3	7.5	0.05474705474705476	1.2616392398030527	EVT-P-0001036-T01-IM3-33, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (34 total)
368	393	25	24	16	22	11	5.0	0.09415068795873749	1.0261765022475087	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (24 total)
293	393	100	28	12	25	8	4.166666666666667	0.16797757509831812	0.7747486924473793	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (28 total)
277	327	50	5	35	3	30	0.25	0.20399751978699343	0.6903751127180751	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0006412-T01-IM5-30, EVT-P-0010945-T01-IM5-5
439	464	25	8	32	8	25	None	0.3087122845636777	0.5104460883278611	EVT-P-0001036-T01-IM3-33, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, EVT-P-0007392-T01-IM5-39, ... (8 total)
193	393	200	29	11	25	8	2.34375	0.36939377457953293	0.43251042803096634	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (29 total)
389	439	50	14	26	13	20	3.9	0.3870387415279055	0.41224556110594196	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (14 total)
338	438	100	25	15	22	11	2.6666666666666665	0.3920474415830453	0.4066613759105142	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (25 total)
356	381	25	19	21	17	16	2.65625	0.4124220374220374	0.3846581366109272	EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, EVT-P-0002305-T01-IM3-31, ... (19 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
302	327	25	4	36	3	30	0.6	0.5522486048801838	0.2578653725929749	EVT-P-0000913-T02-IM5-4, EVT-P-0004477-T01-IM5-3, EVT-P-0005553-T02-IM5-17, EVT-P-0010945-T01-IM5-5
393	418	25	5	35	5	28	None	0.564686143633512	0.24819286892566125	EVT-P-0008567-T01-IM5-38, EVT-P-0008920-T01-IM5-48, EVT-P-0010398-T01-IM5-18, EVT-P-0010494-T01-IM5-15, EVT-P-0011111-T01-IM5-25
393	593	200	15	25	13	20	1.625	0.69127409142889	0.16034971992983757	EVT-P-0001036-T01-IM3-33, EVT-P-0002516-T01-IM3-21, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, ... (15 total)
414	439	25	9	31	8	25	1.92	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (9 total)
388	438	50	6	34	5	28	1.0714285714285714	1.0	-0.0	EVT-P-0003502-T01-IM5-19, EVT-P-0008567-T01-IM5-38, EVT-P-0008920-T01-IM5-48, EVT-P-0010398-T01-IM5-18, EVT-P-0010494-T01-IM5-15, ... (6 total)
281	381	100	24	16	20	13	1.1538461538461537	1.0	-0.0	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (24 total)
238	438	200	30	10	25	8	1.25	1.0	-0.0	EVT-P-0000913-T02-IM5-4, EVT-P-0001687-T01-IM3-41, EVT-P-0001988-T01-IM3-14, EVT-P-0002071-T02-IM5-46, EVT-P-0002197-T01-IM3-44, ... (30 total)
438	638	200	10	30	8	25	0.8	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0002516-T01-IM3-21, EVT-P-0003502-T01-IM5-19, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, ... (10 total)
439	639	200	9	31	8	25	1.92	1.0	-0.0	EVT-P-0001036-T01-IM3-33, EVT-P-0002516-T01-IM3-21, EVT-P-0003664-T01-IM5-29, EVT-P-0004103-T01-IM5-43, EVT-P-0005461-T02-IM5-24, ... (9 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000975 7-T01-IM5-1	P-000975 7-T01-IM5	P-000975 7	Colorectal Cancer	COAD	BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAFinv | Protein fusion: in frame (AGAP3-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000786 1-T01-IM5-2	P-000786 1-T01-IM5	P-000786 1	Non-Small Cell Lung Cancer	LUAD	BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAF | Protein fusion: in frame (AGAP3-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000447-T01-IM5-3	P-000447-T01-IM5	P-000447	Breast Cancer	ILC	BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	327	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFinv | Protein fusion: in frame (AGK-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000913-T02-IM5-4	P-0009113	P-0009113	Non-Small Cell Lung Cancer	LUAD	BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	327	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK _c.981-2119: BRAF inv | Protein fusion: in frame (AGK-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001094 5-T01-IM5-5	P-001094 5-T01-IM5	P-001094 5	Non-Small Cell Lung Cancer	LUAD	BRAF-AGK	AGK	in-frame	three_prime	1404976 10	8	327	True	327	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+12646: AGK_c.980+2552:BRAFinv | Protein fusion: in frame (AGK-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00047955-T01-IM5-13	P-00047955-T01-IM5	P-00047955	Colorectal Cancer	COAD	BRAF-CUL1	CUL1	in-frame	three_prime	140493811	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CUL1 (NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427 :CUL1_c.1140+297: BRAFinv | Protein fusion: in frame (CUL1-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000198 8-T01-IM3-14	P-000198 8-T01-IM3	P-000198 8	Glioma	PAST	BRAF-FAM131B	FAM131B	in-frame	three_prime	1404896 29	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel | Protein fusion: in frame (FAM131B-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001049 4-T01-IM5-15	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-16	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-LINC00244	LINC00244	unknown	five_prime	140492384	8	380	True	380	True	exact match	lost	0.0	False												RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement: c.1140+1724: BRAF_chr7:g.156140327inv | BRAF-NCRNA00244 Fusion	NA
EVT-P-001049 4-T01-IM5-17	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-18	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-19	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-20	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-21	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-22	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-23	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-24	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-25	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-26	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-27	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-28	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-29	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-30	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-31	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-32	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-33	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-34	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-35	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-36	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-37	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-38	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-39	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-40	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-41	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-42	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-43	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-44	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-45	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-46	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-47	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-48	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-49	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame
EVT-P-001049 4-T01-IM5-50	P-001049 4	P-001049 4	Pancreatic Cancer	PAAC	BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655)-BRAF (NM_004333) Rearrangement: t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000553-T02-IM5-17	P-000553-T02-IM5	P-000553	Non-Small Cell Lung Cancer	LUAD	BRAF-MRPS33	MRPS33	unknown	five_prime	140497546	7	327	True	326	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False														RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - MRPS33 (NM_053035)Rearrangement : c.980+2616:BRAF_c.215+390:MRPS33inv | BRAF-MRPS33 Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001039 8-T01-IM5-18	P-001039 8-T01-IM5	P-001039 8	Soft Tissue Sarcoma	MFH	BRAF-OSBPL9	OSBPL9	in-frame	three_prime	140484912	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain			OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement: t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) | Protein fusion: in frame (OSBPL9-BRAF)	in frame
EVT-P-000350 2-T01-IM5-19	P-000350 2	P-000350 2	Non-Small Cell Lung Cancer	LUAD	BRAF-PJA2	PJA2	in-frame	five_prime	140482379	10	438	True	438	True	exact match	lost	0.0	False												RAS-binding domain; Cysteine-rich domain	Protein kinase domain			NA | Protein fusion: in frame (BRAF-PJA2)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000924 6-T01-IM5 -20	P-000924 6-T01-IM5	P-000924 6	Soft Tissue Sarcoma	DDLS	BRAF-RBM33	RBM33	in-frame	three_prime	140482661	11	439	True	439	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAF inv | Protein fusion: in frame (RBM33-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000251 6-T01-IM3 -21	P-000251 6-T01-IM3	P-000251 6	Prostate Cancer	PRAD	BRAF-VIPR2	VIPR2	unknown	five_prime	140454067	13	565	True	564	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.4235294117647059	True	partner_lacks_domain	PF07714						51				False		RAS-binding domain; Cysteine-rich domain		Protein kinase domain	NA | BRAF-VIPR2 Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000583 4-T01-IM5-22	P-000583 4-T01-IM5	P-000583 4	Glioma	PXA	CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	381	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CCDC6 (NM_005436) - BRAF (NM_004333) Rearrangement: t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) | Protein fusion: in frame (CCDC6-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00042033-T01-IM5-23	P-00042033-T01-IM5	P-00042033	Melanoma	SKCM	CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	9	381	True	381	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(7q34;9q33.2)(chr7:140491585::chr9:123251876) | Protein fusion: in frame (CDK5RAP2-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000546 1-T02-IM5-24	P-000546 1-T02-IM5	P-000546 1	Melanoma	SKCM	CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	439	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34;9q33.2)(chr7:140481684::chr9:123255452) | Protein fusion: in frame (CDK5RAP2-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001111-T01-IM5-25	P-001111-T01-IM5	P-001111	Pancreatic Cancer	PAAD	KDM7A-BRAF	KDM7A	unknown	three_prime	140484788	10	393	True	393	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain			BRAF (NM_004333) - JHDM1D (NM_030647) rearrangement: c.1178-1831:BRAF_c.*1672:JHDM1D del | JHDM1D-BRAF Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00090977-T01-IM5-26	P-00090977-T01-IM5	P-00090977	Glioma	PAST	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain) : c.5247+3837: KIAA1549_c.1141-1433:BRAF... (230 chars total)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00403 2-T01-IM5 -27	P-00403 2-T01-IM5	P-00403 2	Glioma	AGNG	KIAA 1549-BRAF	KIAA 1549	in-frame	three_prime	140487892	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA 1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000512 1-T02-IM5-28	P-0005121-T02-IM5	P-0005121	Melanoma	ACRM	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000366 4-T01-IM5-29	P-000366 4-T01-IM5	P-000366 4	Thyroid Cancer	THAP	KIAA1549-BRAF	KIAA1549	in-frame	three_prime	1404828217	11	439	True	439	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain			c.4929+549:KIAA1549_c.1314+4:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000641 2-T01-IM5 -30	P-000641 2	P-000641 2	Pancreatic Cancer	PAAD	LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	286	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False																RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel | Antisense fusion	NA
EVT-P-000230 5-T01-IM3 -31	P-000230 5	P-000230 5	Melanoma	ACRM	MET TL2B-BRAF	MET TL2B	unknown	three_prime	140491674	9	381	True	381	True	exact match	retained	1.0	False													RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | MET TL2B-BRAF Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000902 4-T01-IM5-32	P-0009024	P-0009024	Prostate Cancer	PRAD	MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup | Protein fusion: in frame (MKRN1-BRAF)	in frame
EVT-P-000103 6-T01-IM3-33	P-0001036	P-0001036	Cancer of Unknown Primary	CUP	MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	11	439	True	439	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | Protein fusion: in frame (MKRN1-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000437-3-T02-IM5-34	P-000437-3-T02-IM5	P-000437-3	Germ Cell Tumor	VMGCT	MKRN1-BRAF	MKRN1	unknown	three_prime	140434466	18	744	False	744	True	exact match	lost	0.0	False														Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup | Protein fusion: mid-exon (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000878 9-T01-IM5-36	P-000878 9-T01-IM5	P-000878 9	Melanoma	SKCM	PHTF2-BRAF	PHTF2	in-frame	three_prime	140489236	9	381	True	381	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain			PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367:PHTF2_c.1141-1852:BRAFinv | Protein fusion: in frame (PHTF2-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000969 4-T01-IM5-37	P-000969 4-T01-IM5	P-000969 4	Hepatobiliary Cancer	CHOL	PRKAR1B-BRAF	PRKAR1B	in-frame	three_prime	140491855	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			PRKAR1B (NM_001164758)-BRAF (NM_004333) Rearrangement: c.892-2551:PRKAR1B_c.1140+2253:BRAFdup | Protein fusion: in frame (PRKAR1B-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000856 7-T01-IM5-38	P-000856 7-T01-IM5	P-000856 7	Thyroid Cancer	THPA	PRKAR2B-BRAF	PRKAR2B	in-frame	three_prime	140486416	10	393	True	393	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFinv | Protein fusion: in frame (PRKAR2B-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000739 2-T01-IM5 -39	P-000739 2-T01-IM5	P-000739 2	Thyroid Cancer	THPD	SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	439	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) | Protein fusion: in frame (SCRIB-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000753-T01-IM5-40	P-000753-T01-IM5	P-000753	Prostate Cancer	PRAD	SND1-BRAF	SND1	in-frame	three_prime	140482226	11	439	True	439	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738: SND1_c.1314+595: BRAFinv | Protein fusion: in frame (SND1-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000168 7-T01-IM3 -41	P-000168 7-T01-IM3	P-000168 7	Prostate Cancer	PRSCC	SND1-BRAF	SND1	in-frame	three_prime	1404902 24	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv | Protein fusion: in frame (SND1-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000814 0-T01-IM5-42	P-000814 0-T01-IM5	P-000814 0	Pancreatic Cancer	PAAC	SND1-BRAF	SND1	in-frame	three_prime	140489439	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFinv | Protein fusion: in frame (SND1-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000410 3-T01-IM5-43	P-0041 03	P-0041 03	Non-Small Cell Lung Cancer	LUAD	SND1-BRAF	SND1	in-frame	three_prime	140481685	11	439	True	439	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAF inv. | Protein fusion: in frame (SND1-BRAF)	in frame
EVT-P-000219 7-T01-IM3-44	P-0021 97	P-0021 97	Pancreatic Cancer	PAAC	SND1-BRAF	SND1	in-frame	five_prime	140488303	8	380	True	380	True	exact match	lost	0.0	False												RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SND1_c.1141-922:BRAF inv | Protein fusion: in frame (BRAF-SND1)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000961 9-T01-IM5-45	P-000961 9-T01-IM5	P-000961 9	Pancreatic Cancer	PAAD	SND1-BRAF	SND1	in-frame	three_prime	1404824 94	11	439	True	439	True	exact match	retained	1.0	False													Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFinv | Protein fusion: in frame (SND1-BRAF)	in frame

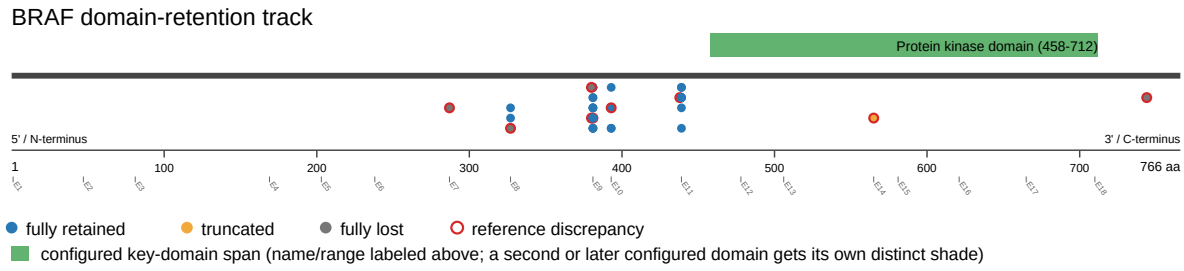
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-00020711-T02-IM5-46	P-00020711-T02-IM5	P-00020711	Pancreatic Cancer	PAAD	SND1-BRAF	SND1	in-frame	three_prime	140489292	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SND1_c.1141-1908:BRAFinv | Protein fusion: in frame (SND1-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001239-T01-IM5-47	P-001239-T01-IM5	P-001239	Pancreatic Cancer	PAAD	SND1-BRAF	SND1	in-frame	three_prime	140493527	9	381	True	381	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFinv | Protein fusion: in frame (SND1-BRAF)	in frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000892-T01-IM5-48	P-000892-T01-IM5	P-000892	Prostate Cancer	PRAD	ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	10	393	True	393	True	exact match	retained	1.0	False												Protein kinase domain	RAS-binding domain; Cysteine-rich domain			ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533)| Protein fusion: in frame (ZNF207-BRAF)	in frame

Figures

domain retention outliers



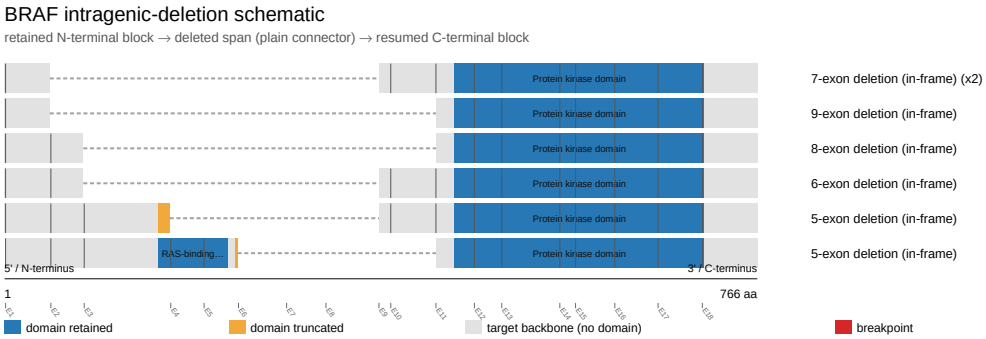
fusion schematic



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 30 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

